

Yaw Nana Gyamfi Prempeh

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PROFESSIONAL SUMMARY

DevSecOps engineer with 6+ years designing, automating and securing cloud infrastructure on AWS, IONOS Cloud, Google Cloud and hybrid environments. Builds Kubernetes clusters from scratch with kubeadm, provisioned through Terraform and Ansible. Writes CI/CD pipelines in GitHub Actions, Jenkins, GitLab CI/CD and Azure DevOps, and runs GitOps with ArgoCD. Secures platforms with HashiCorp Vault, cert-manager and network policies, and monitors them with Prometheus, Grafana and AlertManager. Ingress with Traefik and HAProxy. Mentors engineers and enforces security practice across the DevSecOps toolchain.

TECHNICAL SKILLS

- Container Orchestration: Kubernetes (kubeadm, kubectl, Helm), Docker, Calico CNI, MetalLB, KEDA
- Infrastructure as Code: Terraform, Ansible, CloudFormation, Azure Resource Manager
- CI/CD & GitOps: GitHub Actions, Jenkins, GitLab CI/CD, Azure DevOps Pipelines, ArgoCD
- Cloud Platforms: AWS (EC2, S3, IAM, Lambda, EKS), IONOS Cloud, Google Cloud, Heroku, Azure
- Monitoring & Observability: Prometheus, Grafana, AlertManager, Loki, Promtail, ELK Stack
- Security & Secrets Management: HashiCorp Vault, External Secrets Operator, cert-manager, RBAC, Network Policies
- Ingress & Networking: Traefik, HAProxy, Nginx, Apache, Let's Encrypt, TLS/SSL
- Programming & Scripting: Python, Bash, JavaScript/TypeScript, Node.js
- Databases: PostgreSQL, MySQL, MongoDB, Redis
- Operating Systems: Linux (Ubuntu, CentOS, RHEL), Windows Server

PROFESSIONAL EXPERIENCE

DevSecOps Engineer (Contract)

Bosonit – Spain (Hybrid/Remote from Ghana)

November 2025 – January 2026

- Architected and deployed production-grade Kubernetes infrastructure from scratch on IONOS Cloud using Terraform and Ansible for the ECDC (European Centre for Disease Prevention and Control) compliance platform. Nodes sit on a private network, reachable only through an OpenVPN bastion.
- Designed and implemented a 7-stage Azure DevOps CI/CD pipeline automating infrastructure provisioning, cluster bootstrap, and application deployment.
- Deployed comprehensive DevOps toolchain including HashiCorp Vault for secrets management, ArgoCD for GitOps, Jenkins for CI/CD, and Prometheus/Grafana monitoring stack with custom dashboards and alerting.
- Configured Traefik as the ingress controller with MetalLB load balancing, and HAProxy on the bastion host as a reverse proxy to the control plane and worker nodes, consolidating multiple service endpoints into a single entry point.
- Implemented cert-manager with a self-signed CA for automated TLS certificate provisioning across cluster services.
- Deployed and configured Eclipse Tractus-X EDC (Eclipse Dataspace Connector) for secure data exchange, including Helm chart customisation and Vault integration.
- Led local deployment and validation of Gaia-X Digital Clearing House (GXDCH) Loire services, creating API test suites for compliance and registry services, with 9 of 12 compliance tests passing.
- Authored 80+ pages of technical documentation including setup guides, architecture diagrams, security best practices, and runbooks for team knowledge transfer.

Lead DevOps Engineer (Part-Time)

Algo AI – Canada (Remote)

August 2025 – October 2025

- Architected and automated CI/CD pipelines with GitHub Actions, minimising manual deployment errors.
- Deployed and managed six full-stack applications (Vue.js, TypeScript, Tailwind, Node.js, Python, FastAPI microservices) on Heroku.
- Optimised frontend and backend codebases to reduce build failures and speed up release cycles.
- Implemented application monitoring and alerting with notifications at an 80% threshold.
- Used Heroku Scheduler to scale dynos on a schedule and cut hosting costs.
- Deployed Docker-based containers on Heroku to reduce environment-specific issues.

DevOps Engineer

Technology Excellence Services – USA (Remote)

May 2022 – March 2026

- Automated build, test, and deployment workflows using Jenkins and GitLab CI/CD, improving release consistency.
- Monitored and diagnosed system performance and resource use with Prometheus and Grafana.
- Automated infrastructure provisioning with Terraform and Ansible, ensuring reproducible deployments across development, staging, and production.
- Implemented security best practices including access controls, vulnerability management, and compliance checks.
- Managed version control, branching strategies, and deployment history with Git, improving traceability and rollback capabilities.

Platform Infrastructure Engineer

Blue Turtle Technologies – South Africa (Remote)

May 2020 – April 2022

- Designed and maintained scalable cloud infrastructure, improving system reliability.
- Provisioned and managed cloud resources across AWS and Google Cloud, optimising usage and monthly costs.
- Implemented Infrastructure-as-Code using Terraform and CloudFormation, replacing manual provisioning.
- Managed security measures including access controls, encryption, and threat detection, strengthening compliance.
- Provided technical guidance and mentorship to team members, fostering a collaborative environment.

Linux System Engineer

Millicom Tigo – Ghana

February 2019 – March 2020

- Managed Linux-based servers (Ubuntu, CentOS, RHEL) for performance, security, and availability.
- Installed, configured, and troubleshot Linux systems and services (Apache, Nginx, MySQL).
- Implemented security measures including firewalls and intrusion detection systems, reducing vulnerabilities.
- Documented system configurations and procedures, ensuring knowledge sharing and faster onboarding.

PROJECTS

K8s Troubleshoot MCP github.com/NanaGyamfiPrempeh30/k8s-troubleshoot-mcp

- Built a read-only MCP (Model Context Protocol) server exposing 16 diagnostic tools that let AI assistants (Claude Desktop, VS Code, Kiro) diagnose Kubernetes clusters via natural language instead of manual kubectl correlation.
- Designed an RBAC-enforced security boundary (scoped ServiceAccounts, zero write verbs, automated can-i checks) plus structured output escaping to mitigate prompt injection from untrusted cluster log data.
- Validated with 453 passing tests, including 18 Hypothesis property suites, then verified against a live cluster; packaged as a non-root Docker image (UID 10001) with TruffleHog secret scanning in CI.

Nextcloud Installer github.com/NanaGyamfiPrempeh30/nextcloud-installer

- Built a fully unattended, idempotent CLI installer for Nextcloud on native Linux (nginx, PHP-FPM, MariaDB, Redis), replacing Nextcloud's web-based setup wizard with scriptable occ CLI automation.
- Designed a staged pipeline with per-category exit codes and a three-state install-check model to prevent destructive reinstalls on transient failures (e.g. database unreachable).
- Verified end-to-end on Ubuntu 24.04 LTS, including failure recovery, idempotent re-runs, and ONLYOFFICE Document Server integration.

Ambient Weather MCP github.com/NanaGyamfiPrempeh30/ambient-weather-mcp

- Built and published an MCP server connecting AI assistants to personal weather station data via natural language queries, developed using a spec-driven workflow with mentorship from senior DevSecOps and AI leader Wilson Mar.
- Implemented a 60-second TTL cache to stay within strict third-party API rate limits, and shipped a GitHub Actions CI/CD pipeline publishing to GitHub Container Registry with TruffleHog secret scanning.
- Deployed via Kubernetes manifests for ArgoCD-managed GitOps delivery.

Sika Track github.com/NanaGyamfiPrempeh30/Sika-Track

- Built a Telegram bookkeeping bot for informal businesses in Ghana, letting owners log daily sales and expenses through plain chat messages (e.g. "sold 50", "spent 30 gas") instead of manual paper records.
- Implemented category-aware message parsing and on-demand daily/weekly profit summaries; built with Flask and a webhook-based Telegram Bot API integration, containerised with Docker for deployment.

EDUCATION

Bachelor of Science in Information Technology

Ghana Communication Technology College – Tesano, Accra | 2012 – 2015

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (2024)
- AWS Certified Cloud Practitioner (2024)
- AWS Technical Essentials (course, 2024)

LANGUAGES

English: Fluent

REFERENCES

Available on request.